Global-Group Attention Network With Focal Attention Loss for Aerial Scene Classification

Yichen Zhao, Yaxiong Chen, Yi Rong, Shengwu Xiong, and Xiaoqiang Lu, *Senior Member, IEEE*

***Abstract*— Aerial scene classification, aiming at assigning a specific semantic class to each aerial image, is a fundamental task in the remote sensing community. Aerial scene images have more diverse and complex geological features. While some statistics of images can be well fit using convolution, it limits such models to capturing the global context hidden in aerial scenes. Furthermore, to optimize the feature space, many methods add class information to the feature embedding space. However, they seldom combine model structure with class information to obtain more separable feature representations. In this article, we propose to address these limitations in a unified framework (i.e., CGFNet) from two aspects: focusing on the key information of input images and optimizing the feature space. Specifically, we propose a global-group attention module (GGAM) to adaptively learn and selectively focus on important information from input images. GGAM consists of two parallel branches: the adaptive global attention branch (AGAB) and the region-aware attention branch (RAAB). AGAB utilizes an adaptive pooling operation to better model the global context in aerial scenes. As a supplement to AGAB, RAAB combines grouping features with spatial attention to spatially enhance the semantic distribution of features (i.e., selectively focus on effective regions of features and ignore irrelevant semantic regions). In parallel, a focal attention loss (FA-Loss) is exploited to introduce class information into attention vector space, which can improve intraclass consistency and interclass separability. Experimental results on four publicly available and challenging datasets demonstrate the effectiveness of our method. The source code will be released at: https://github.com/zoecheno/CGFNet.**

***Index Terms*— Aerial scene classification, attention, convolutional neural networks (CNNs), loss function, remote sensing.**

Manuscript received 15 July 2023; revised 1 October 2023; accepted 22 November 2023. Date of publication 13 December 2023; date of current version 20 December 2023. This work was supported in part by the NSFC under Grant 62101393; in part by the Project of Sanya Yazhou Bay Science and Technology City under Grant SCKJ-JYRC-2022-17; in part by NSFC under Grant 62176194; in part by the Project of Sanya Yazhou Bay Science and Technology City under Grant SKJC-2022-PTDX-031; in part by the National Key Research and Development Program of China under Grant 2022ZD0160604; in part by the Key Research and Development Program of Hubei Province under Grant 2023BAB083; in part by the Hainan Province "Nanhai New Star" Technology Innovation Talent Platform Project under Grant NHXXRCXM202361; in part by the Youth Fund Project of Hainan Natural Science Foundation under Grant 6220N344; in part by the Chinese Association for Artificial Intelligence (CAAI)-Huawei MindSpore Open Fund under Grant CAAIXSJLJJ-2022-001A; in part by the Knowledge Innovation Program of Wuhan-Basic Research; in part by the Sanya Science and Education Innovation Park of Wuhan University of Technology under Grant 2022KF0020; in part by the High-Performance Computing Platform of YZBSTCACC; in part by the Natural Science Foundation of Chongqing under Grant cstc2021jcyj-msxmX1148; and in part by MindSpore, which is a new deep learning computing framework. *(Corresponding authors: Yaxiong Chen; Shengwu Xiong.)*

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Digital Object Identifier 10.1109/TGRS.2023.3342171

I. Introduction

As A fundamental task in the remote sensing community, aerial scene classification is rapidly becoming a crucial player in many practical applications, such as urban planning, land management, and natural hazard detection [1], [2]. The goal is to match each aerial image with a reasonable semantic class, e.g., Park, Forest, and Beach [3]. Aerial images contain diverse and complex information about the structure of the Earth's surface. The challenge of aerial scene classification mainly lies in extracting effective feature representations from images (for identifying the same class and distinguishing different classes).

In recent years, advances in convolutional neural networks (CNNs) have greatly facilitated the remote sensing community. CNNs have powerful nonlinear fitting ability, which can extract high-level visual features from images. Therefore, many improved CNN-based methods have emerged for improving the accuracy of aerial scene classification. CNNs usually contain millions of parameters and require large amounts of data to drive training. Most CNN-based methods directly use a CNN (e.g., ResNet [5]) pretrained on ImageNet [6] as their backbone network and further improve it [7], [8], [9]. However, two key challenges remain as follows.

On one hand, while using convolutions can well fit some statistics of images, it may also prevent such models from capturing the global context. Concretely, the global context is essential for intelligent image interpretation, which refers to information captured over the entire spatial extent of image features. With continuous stacking of convolutional layers, the receptive field of many CNN-based methods is large enough to completely cover the input image. However, Luo et al. [10] shows that the effective receptive field of CNNs is actually much narrower than their theoretical receptive field. Moreover, the global context can provide clues to local ambiguities, which is extremely helpful for CNNs to correctly classify aerial scene images with multiple geological features. As shown in Fig. 1, both the class "palace" and the class "dense residential" contain the same land covers, such as houses and plants. The difference between the two is mainly reflected in the overall spatial distribution. Therefore, exploiting the global context explicitly can often improve the performance of CNNs for aerial scene classification. Many studies resort to attention mechanisms [11], [12], [13], [14].

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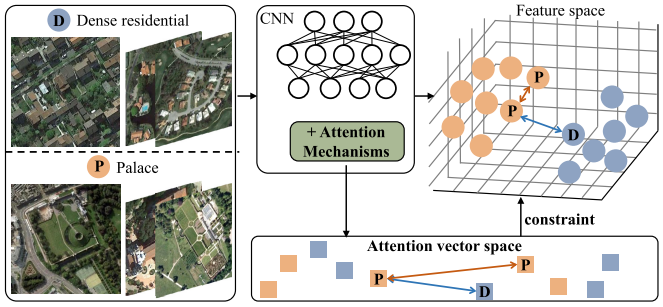


Fig. 1. Above aerial scene images are selected from the NWPU RESICS-45 dataset [4]. 1) It can be seen that both the class "palace" and the class "dense residential" contain the same land covers, such as houses and plants. The difference between the two is mainly reflected in the overall spatial distribution. Therefore, both global and local information should be considered. 2) The attention vector space in attention-based methods provides us with a space to further constrain the class relationship between samples. Thus, the feature space can be optimized by introducing attention mechanisms and enforcing class relation constraints in channel attention.

As a pioneer in channel attention, Hu et al. [11] obtains the global context through a global avg-pooling operation and uses this information to recalibrate the channel feature response. GENet [12] is designed to gather global contextual information by reweighting on the spatial domain. In addition, vision transformers (ViTs) have also attracted scholars' favor due to their effectiveness in learning global contextual information [13], [15], [16], [17]. For example, EMTCAL [13] takes advantage of CNN in local modeling and ViT in global modeling to mine information within aerial scenes fully. Unlike CNNs, the large computational load of ViTs has always constrained their development in the field of remote sensing [15].

On the other hand, most aerial scene classification methods apply cross-entropy loss (CE-Loss) to supervise the training of the model. However, CE-Loss is insufficient to guide the model to extract more discriminative features from aerial scene images with complex backgrounds. To be specific, CE-Loss measures the distance between two probability distributions. Classification accuracy is improved by increasing the predicted softmax probability corresponding to the true label. Nonetheless, the visual features extracted by models trained with CE-Loss are separable between classes but loosely distributed within classes, lacking sufficient discriminative power [18]. The above phenomenon weakens generalization performance and impairs model performance on vision tasks. To address this limitation, many improved loss functions add class information to the feature embedding space [19], [20], [21]. However, they seldom consider combining the model structure with class information to extract more discriminative features. Thanks to the projection head in SimCLR [22], feature representations are mapped to the latent space to compute contrastive loss. Chang et al. [23] acts directly on the feature channel, encouraging a set of channels to be class-aligned and each to be discriminative on mutually distinct local regions. Notably, the attention vector space in attention-based methods inherently provides a space to further constrain the class relationship between samples. DNNs encode images into feature representations. After that, feature representations from images of the same class are clustered in a high-dimensional feature embedding space [5], [22]. Similarly, since attention vectors are a map of feature importance [11], [24], channel attention vectors of the same class should be more clustered in the vector space than those of different classes. This inspires us to encourage the model to learn a more reasonable feature space by describing the similarity between samples in the attention vector space. Thus, the aerial scene classification task can benefit from it.

In this article, we propose to address these limitations in a unified framework (i.e., CGFNet) from two aspects: focusing on the key information of the input image and optimizing the feature space. To be specific, a global-group attention module (GGAM) is developed to adaptively learn and selectively focus on important information from input images. GGAM is inserted into each residual unit of ResNet50 [5], which includes two parallel branches: the adaptive global attention branch (AGAB) and the region-aware attention branch (RAAB). Given an extracted feature map, GGAM employs AGAB and RAAB to generate an attention map in channel and spatial dimensions, respectively. To enhance informative channels and important regions, the feature maps reweighted by the attention map are added element-wise for fusion. Particularly, AGAB utilizes an adaptive pooling operation to better model the global context in aerial scenes. This global information is then used to recalibrate the weight of each channel. As a supplement to AGAB, RAAB combines grouping features with spatial attention to spatially enhance the semantic distribution of features (i.e., selectively focus on effective regions of features and ignore irrelevant semantic regions). In parallel with GGAM, a focal attention loss (FA-Loss) is exploited to encourage channel attention vectors from samples of the same class to be close together, while expanding the distance of channel attention vectors between samples of different classes. In this way, label information can play a more active role in model training.

In summary, this article includes fourfold contributions.

1) A novel method for accurate aerial scene classification called CGFNet is proposed. It can extract more discriminative features hidden in aerial scenes and eliminate irrelevant information.
2) The GGAM is proposed to adaptively learn and selectively focus on important information of input images.
3) A novel FA-Loss function is proposed. It introduces class information into attention vector space, which can improve intraclass consistency and interclass separability.
4) Experiments on four open datasets demonstrate the effectiveness of our method for aerial scene classification.

The remainder of this article is organized as follows. Section II briefly reviews existing methods concerned with our work. Section III elaborates the detailed process of our proposed CGFNet. Section IV exhibits the experimental results and analyses. Finally, Section V draws the conclusion.

II. Related Works

A. Attention-Based Methods for Aerial Scene Classification

Attention mechanisms modify the model's attention to the most important regions in the image and ignore irrelevant

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parts. They are beneficial in many visual tasks, such as image classification [11], [12], [24], and few-shot learning [25], [26]. Many attention mechanisms explicitly take global context into account. As a pioneer in channel attention, SENet [11] proposed a squeeze and excite (SE) block for collecting the global context and improving representation ability. Inspired by SENet, Gather-Excite (GE) [12] is designed to gather global contextual information by reweighting on the spatial domain. Channel attention adaptively reweights each channel, which can be seen as a feature screening process. Relatedly, spatial attention can be viewed as a process of screening spatial regions. There are also some methods dedicated to decoupling attention from channels and spaces to further enhance the representation ability of CNNs. BAM [27] and CBAM [24] reweigh in channel and spatial dimensions, respectively, to secure more refined feature maps. To reduce additional computational effort, features are divided into groups of subfeatures by SA-Net [28] along the channel dimension. Then, SA-Net computes channel and spatial attention for each subfeature separately and finally utilizes a shuffle unit for integration. Li et al. [29] proposes a spatial group-wise enhance (SGE) module to autonomously enhance the feature representation of each subfeature group and suppress possible noise.

Tons of methods have demonstrated excellent performance in aerial scene classification using CNNs. However, they are not good at extracting key features from complex backgrounds and are easily disturbed by irrelevant information. Many studies resort to attention mechanisms, which improve the performance of CNNs by highlighting pertinent information in an image and filtering redundant information [13], [14], [24]. Chen et al. [14] proposed the method named MBLANet, which embeds the newly devised module (CLAM) into the ResNet50 network. CLAM can simultaneously obtain channel attention and spatial attention, which highlights main regions in complex backgrounds and enriches feature representations. With the emergence of ViT [15], a host of ViT-based methods have been proposed and achieved comparable improvements to CNNs in aerial scene classification. Tang et al. [13] proposed a model named EMTCAL, which utilizes the strengths of CNN in local modeling and ViT in global modeling to mine information within aerial scenes fully. In this work, we propose a GGAM for aerial scene classification. GGAM consists of two parallel branches: the AGAB and the RAAB. AGAB utilizes an adaptive pooling operation to better model the global context in aerial scenes. As a supplement to AGAB, RAAB combines grouping features with spatial attention to spatially enhance the semantic distribution of features (i.e., selectively focus on effective regions of features and ignore irrelevant semantic regions). With the help of GGAM, the feature representation capability of our baseline can be effectively improved.

B. Loss Functions in CNNs

A tendency to design loss functions for specific objectives has been noticed in various computer vision tasks. Current prevailing classification loss functions are mainly based on CE-Loss, which guides the model by narrowing the distance between the true label and the predicted softmax probability distribution. CE-loss is skilled in ameliorating interclass discrimination (i.e., separating features from different classes), but is unable to maintain intraclass consistency (i.e., compacting features from the same class). Iqbal et al. [18] constrains the distance of features to their class centers by adding a regularization term. To push the class boundaries closer to each class's weight vector, a conceptually attractive angular margin is introduced by Liu et al. [19] and Wang et al. [20]. Furthermore, the arcface loss [21] maximizes directly the classification boundary in the angular space. There are also some loss functions that focus on hard negative mining to improve model performance. The classic approach is focal loss [30], which is a dynamically scaled CE-Loss. More specifically, the focal loss dynamically decreases the weight of easy samples during training by a dynamic scaling factor, resulting in the weight quickly on hard samples.

However, the losses mentioned above rarely consider combining the model structure with class information to extract more discriminative features. Through the projection head in SimCLR [22], feature representations are mapped to the latent space to compute the contrastive loss. Chang et al. [23] attempts to learn and localize local discriminative features with only a loss without introducing any explicit network components. It acts directly on the feature channel, encouraging a set of channels to be class-aligned and each to be discriminative on mutually distinct local regions. Inspired by MC-Loss, Song et al. [31] proposed channel correlation loss (CC-Loss). CC-Loss obtains more discriminative features by encouraging different channels of features corresponding to their classes. The attention vector space in attention-based methods provides us with a space to further constrain the class relationship between samples. Describe the similarity between samples in the attention vector space, and then encourage the model to learn feature representations with larger interclass distances and smaller intraclass distances. In this article, we propose a novel FA-Loss to extract more discriminative features. It introduces class information into attention vector space, which can improve intraclass consistency and interclass separability.

III. Proposed Method

The overall structure of the proposed CGFNet is illustrated in Fig. 2, which consists of two main parts: a GGAM and a novel FA-Loss function. We use ResNet50 [5] as the backbone network, which consists of five stages. In the process of learning feature representation for aerial scene images, CGFNet uses GGAM to purify the feature space and FA-Loss to optimize the feature space. Concretely, GGAM is inserted into each residual unit of ResNet50, aiming to adaptively learn and selectively focus on important information of input images. FA-Loss introduces class information into attention vector space, which can improve intraclass consistency and interclass separability. More implementation details are as follows.

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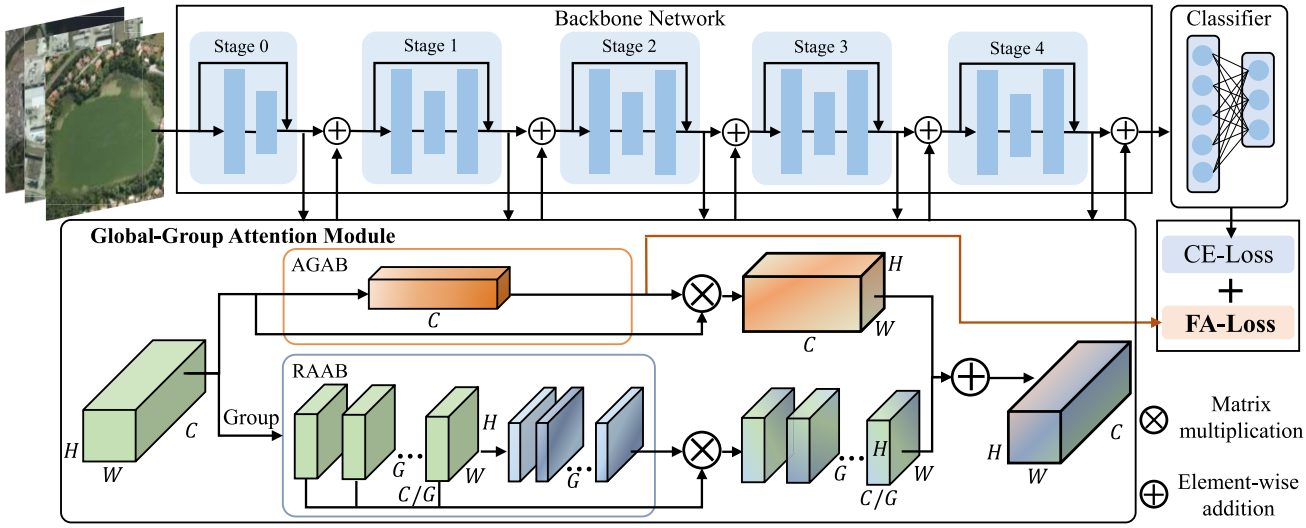


Fig. 2. Framework of CGFNet, which consists of two main parts: a GGAM and a FA-Loss. We use ResNet50 [5] as the backbone network, which consists of five stages. Specifically, GGAM is inserted into each residual unit of ResNet50, which includes two parallel branches: the AGAB and the RAAB. Given an extracted feature map, GGAM employs AGAB and RAAB to generate an attention map in the channel and spatial axis, respectively. Subsequently, the feature maps reweighted by the attention map are added element-wise for fusion to enhance informative channels and effective regions. FA-Loss encourages channel attention vectors from samples of the same class to be close together, while expanding the distance of channel attention vectors between samples of different classes. Thus, the model's ability to extract discriminative features can be further improved.

A. Global-Group Attention Module

GGAM consists of two parallel branches: the AGAB and the RAAB. AGAB utilizes an adaptive pooling operation to better model the global context in aerial scenes, and then captures interchannel dependencies to recalibrate the weights of individual channels. As a supplement to AGAB, RAAB combines grouping features with spatial attention to spatially enhance the semantic distribution of features. Thus, the model can selectively focus on effective regions of features and ignore irrelevant semantic regions. The details of the two branches are as follows.

1) Adaptive Global Attention Branch: Each channel in a feature map usually contains a specific feature response. Channel attention can adaptively reweight each channel, which can be seen as a feature screening process. Most current channel attention mechanisms follow the same paradigm [11]: a squeeze operator and an excite operator. The squeeze operator is designed to aggregate the global context in the spatial dimension, i.e., each 2-D feature map is represented by a real number. The excite operator is designed to capture channel-wise relationships and output the final channel attention map. Global avg-pooling is often used as the squeeze operator to model the global context, which is insufficient for remote sensing images with complex geological features. To better model the global context in aerial scenes, the proposed AGAB utilizes an adaptive pooling operation to adaptively aggregate the global context and uses this global information to recalibrate the weight of each channel.

As shown in Fig. 3, the proposed AGAB consists of two parts: a squeeze operator ξ_S and an excite operator ξ_E. To adaptively collect the global contextual information from feature maps, ξ_S performs an adaptive pooling operation. The excite operator ξ_E is designed as a 1×1 convolutional subnetwork to settle the interaction between channels. More

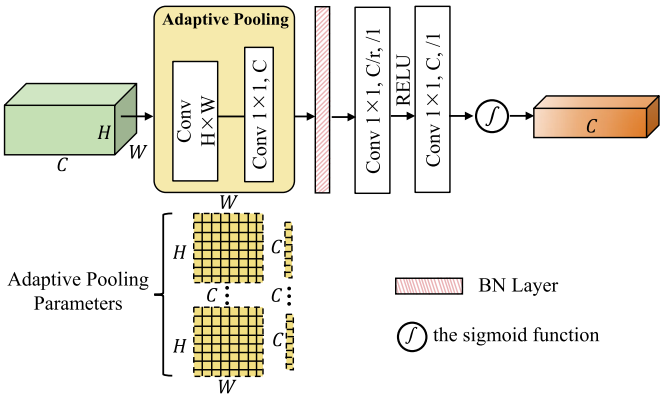


Fig. 3. Detailed structure diagram of the AGAB. To better model the global context in aerial scenes, an adaptive pooling operation is used to adaptively aggregate global contextual information. This global information is then used to recalibrate the weights of each channel.

formally, for a feature map $\mathbf{F}$ as input, its number of channels, spatial height, and width are C, H, and W, respectively. ξ_S performs an adaptive pooling operation on $\mathbf{F}$ and produces a channel descriptor $\mathbf{F}_c \in \mathbb{R}^{C\times1\times1}$. In this article, we leverage global depthwise convolution for the adaptive pooling. Then, $\mathbf{F}_c$ is forwarded to the 1×1 convolutional subnetwork ξ_E (ξ_E consists of one 1×1 convolution, one ReLU, one 1×1 convolution, and a sigmoid function in turn) to capture channel-wise dependencies and output the channel attention map $\mathbf{M}_C(\mathbf{F}) \in \mathbb{R}^{C\times1\times1}$. To save on parameter overhead, the output size of the middle layer in ξ_E is set to $C/r \times 1 \times 1$, where r is the reduction ratio. In short, AGAB can take the form

$$\begin{aligned}\mathbf{M}_C(\mathbf{F}) &= \xi_E(\xi_S(\mathbf{F})) \\ &= \sigma\left(f^{1\times1}\left(\mathrm{ReLU}\left(f^{1\times1}\left(\mathrm{BN}\left(f^{G}(\mathbf{F})\right)\right)\right)\right)\right) \end{aligned} \quad (1)$$

where $f^{1\times1}$ represents a convolution operation with a filter size of 1×1, and f^G indicates the adaptive pooling operation.

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σ is the sigmoid activation function. ReLU(·) is the ReLU activation function. BN(·) denotes the batch normalization technique.

2) Region-Aware Attention Branch: As a supplement to AGAB, RAAB combines grouping features with spatial attention to spatially enhance the semantic distribution of features (i.e., selectively focus on effective regions of features and ignore irrelevant semantic regions). The idea of grouping features has been widely used in CNNs [28], [29], [32], [33]. Some of them group features into multiple subfeatures along the channel, and argue that each subfeature group can capture specific semantic information [29], [32]. In addition, spatial attention can be viewed as a process of screening spatial regions to decide where to focus. Accordingly, RAAB applies spatial attention to capture different semantic information about each feature group.

More specifically, for a given feature map $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$, $\mathbf{F}$ is separated into G groups on the channel. That is, $\mathbf{F} = \left[\mathbf{F}^1, \ldots, \mathbf{F}^g, \ldots, \mathbf{F}^G\right]$, $\mathbf{F}^g \in \mathbb{R}^{(C/G)\times H\times W}$, in which a specific semantic response are gradually captured by each subfeature $\mathbf{F}^g$ during training. Then, the spatial relationship of features is exploited by RAAB to generate corresponding spatial attention for each subfeature. As illustrated in Fig. 4, given a sub-feature $\mathbf{F}^g$, RAAB uses global avg-pooling and global max-pooling operations in parallel on the channel dimension and concatenates them to build an efficient spatial descriptor $\mathbf{F}_s$. Immediately afterward, a 3×3 convolution is applied to $\mathbf{F}_s$. In order to prevent the biased magnitude of coefficients among different samples, each group introduces a pair of parameters to shift the weight values of different groups, which has been widely used [29], [34], [35]. The number of parameters is around tens (64 in this article), which is basically negligible compared to the millions of parameters in the entire model. Finally, $\mathbf{F}_s$ is activated by the sigmoid function to produce the spatial attention map $\mathbf{M}_S(\mathbf{F}) \in \mathbb{R}^{1\times H\times W}$. In short, RAAB can take the form

$$\mathbf{M}_S(\mathbf{F}) = \sigma\left(f^{3\times3}\left(\left[\mathrm{GAP}^c(\mathbf{F}); \mathrm{GMP}^c(\mathbf{F})\right]\right)\right) \tag{2}$$

where $f^{3\times3}$ indicates a convolution operation with a filter size of 3×3. GAP^c is the global average pooling operation on the channel axis. GMP^c is the global max pooling operation on the channel axis. $[\cdot]$ denotes concatenation over channels. σ represents the sigmoid function.

3) Combination of AGAB and RAAB: For an input feature map $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$, AGAB derives a channel attention map $\mathbf{M}_C(\mathbf{F}) \in \mathbb{R}^{C\times1\times1}$, while RAAB derives a spatial attention map $\mathbf{M}_S(\mathbf{F}) \in \mathbb{R}^{1\times H\times W}$. Each attention map is first acted upon individually on the original feature to obtain two refined feature maps. Then, we fuse the two features using an element-wise sum. Finally, a novel attention module, named the GGAM, is achieved to improve the feature representation ability of our baseline for aerial images. GGAM can adaptively select both important feature channels and regions using AGAB and RAAB.

For the combination of the two attention branches, our experimental results show that the parallel structure performs best in aerial scene classification. A detailed discussion and experimental results can be found in Section IV-D. At last, the refined feature map $\mathbf{F}'$ is computed as

$$\mathbf{F}' = \mathbf{F} + (\mathbf{F} \times \mathbf{M}_C(\mathbf{F}) + \mathbf{F} \times \mathbf{M}_S(\mathbf{F}))/2. \tag{3}$$

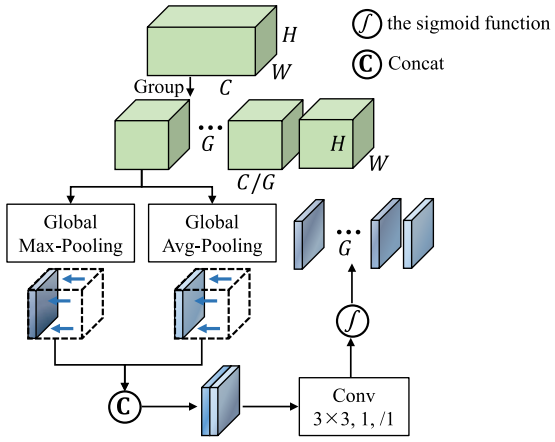


Fig. 4. Detailed structure diagram of the RAAB. RAAB combines grouping features with spatial attention to spatially enhance the semantic distribution of features (i.e., selectively focus on effective regions of features and ignore irrelevant semantic regions). For a given feature map $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$, $\mathbf{F}$ is separated into G groups on the channel. Then, RAAB uses two pooling operations along the channel axis and integrates them to build an efficient spatial descriptor. Finally, a convolution and the sigmoid function are applied to produce the spatial attention map.

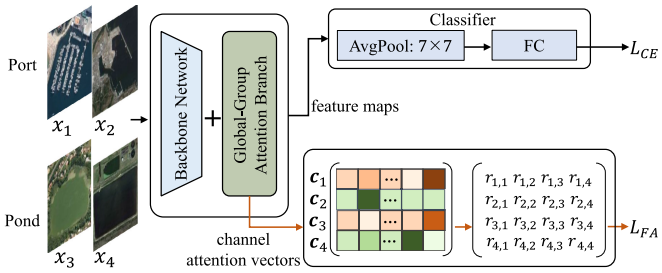


Fig. 5. Pipeline of the proposed FA-Loss. To promote label information to play an active role in feature learning, we utilize sample labels to indicate the class relationship between samples. Concretely, samples of the same class form positive sample pairs and samples of different classes form negative sample pairs. FA-Loss simultaneously pulls positive pairs closer and negative pairs farther away to guide the model to extract more discriminative features.

B. Focal Attention Loss

The attention vector space in attention-based methods provides us with the opportunity to further constrain the class relationship between samples. Specifically, feature representations from images of the same class are clustered in a high-dimensional feature embedding space. The attention vector reflects the important information of the feature space. Naturally, channel attention vectors of the same class should be more clustered in the vector space than those of different classes. We can encourage the model to learn feature representations with larger interclass distances and smaller intraclass distances by describing the similarities between samples in the attention vector space. Therefore, a novel FA-Loss is proposed.

In FA-Loss, a mini-batch of channel attention vectors is used to represent intraclass and interclass relations. The channel attention vector for each sample can be conveniently obtained from AGAB. Euclidean distance is employed to measure the

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distance between different channel attention vectors. Additionally, the labels of samples are used directly to form positive and negative sample pairs. FA-Loss simultaneously pulls positive pairs closer and negative pairs farther away. Fig. 5 shows the FA-Loss processing flow. Mathematically, for a set of N randomly sampled sample/label pairs $\{x_i, y_i\}_{i=1}^{N}$, the CE-Loss in each mini-batch can be defined as

$$L_{\mathrm{CE}} = -\frac{1}{N}\sum_{i=1}^{N} y_i \log(p_i) \tag{4}$$

where p_i is the probability distribution output by the classifier.

A channel attention vector can be represented as $\boldsymbol{c} = \{c_1, c_2, \ldots, c_C\}$, where C is the channel number of features. After that, the channel attention vectors $\mathbf{T}_{\boldsymbol{c}}$ from one mini-batch can be expressed as $\mathbf{T}_{\boldsymbol{c}} = [\boldsymbol{c}_1, \boldsymbol{c}_2, \ldots, \boldsymbol{c}_N]$. We first calculate the Euclidean distance between channel attentions in a mini-batch, resulting in the matrix $\mathbf{D}_{\boldsymbol{c}}$ as follows:

$$\mathbf{D}_{\boldsymbol{c}} = \begin{bmatrix} r_{1,1} & r_{1,2} & \cdots & r_{1,N} \\ r_{2,1} & r_{2,2} & \cdots & r_{2,N} \\ \vdots & \vdots & \ddots & \vdots \\ r_{N,1} & r_{N,2} & \cdots & r_{N,N} \end{bmatrix} \tag{5}$$

where $r_{i,j} = \left|\left|\boldsymbol{c}_i - \boldsymbol{c}_j\right|\right|_2^2$ is the Euclidean distance between the channel attention vector of ith sample and jth sample. Due to the symmetry of the Euclidean distance, it is obvious that $r_{i,j} = 0$ and $r_{i,j} = r_{j,i}$. Consequently, $\mathbf{D}_{\boldsymbol{c}}$ can be further simplified to the following upper triangular matrix:

$$\mathbf{D}_{\boldsymbol{c}}^{\mathrm{UT}} = \begin{bmatrix} 0 & r_{1,2} & \cdots & r_{1,N} \\ 0 & 0 & \cdots & r_{2,N} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}. \tag{6}$$

Labels can be used directly to show positive and negative relationships between samples. To put it simply, samples of the same class form positive sample pairs, and samples of different classes form negative sample pairs. Excluding extraneous factors, FA-Loss only optimizes the average distance between positive and negative sample pairs. It can be formulated as

$$L_{\mathrm{FA}} = L_{\mathrm{intra}}/(L_{\mathrm{inter}} + \eta) \tag{7}$$

with

$$\begin{aligned} L_{\mathrm{intra}} &= \sum_{i=1}^{N} L_{\mathrm{intra}}^{i} \\ &= \frac{1}{|S_i|}\sum_{i=1}^{N}\sum_{j=1}^{N} \mathbb{I}\left(y_j \in S_i\right) r_{i,j} \end{aligned} \tag{8}$$

$$\begin{aligned} L_{\mathrm{inter}} &= \sum_{i=1}^{N} L_{\mathrm{inter}}^{i} \\ &= \frac{1}{N-|S_i|}\sum_{i=1}^{N}\sum_{j=1}^{N} \mathbb{I}\left(y_j \notin S_i\right) r_{i,j} \end{aligned} \tag{9}$$

where η is a small positive constant to avoid zero in the denominator. $|S_i|$ is the size of S_i, and S_i indicates the positive sample set for ith sample. $\mathbb{I}(\cdot) \in \{0, 1\}$ is an indicator function. It outputs 1 when the input condition is true, otherwise it outputs 0.

Finally, the following formula can be obtained as the overall loss function:

$$L = L_{\mathrm{CE}} + \beta \times L_{\mathrm{FA}} \tag{10}$$

where β is a hyperparameter to trade-off L_{CE} and L_{FA}.

IV. Experiments

A. Datasets Description

Four publicly available datasets are used to validate our method for aerial scene classification, which are the UC-Merced Land Use dataset (UCM) [3], the aerial image dataset (AID) [1], the NWPU RESICS-45 dataset [4], and the OPTIMAL-31 dataset [2]. The following briefly lists the information of each dataset.

1) UC-Merced Land Use Dataset (UCM): UCM [3] contains 2100 aerial scene images, each with a spatial resolution of one foot. 2100 images with 256 × 256 pixels are equally carved up into 21 scene classes.

2) Aerial Image Dataset: AID [1] consists of 10 000 images and 30 classes in RGB color space. The number of images for each class ranges from 220 to 420. Each image has a spatial resolution ranging from about 8 to 0.5 m and is 600 × 600 pixels.

3) NWPU RESICS-45 Dataset (NWPU): NWPU [4] contains 31 500 aerial images classified into 45 classes. Each class contains 700 images with spatial resolutions ranging from 30 to 0.2 m and 256 × 256 pixels. NWPU is a large-scale open dataset, which is also one of the most popular big datasets.

4) OPTIMAL-31 Dataset (OPTIMAL-31): OPTIMAL-31 [2] consists of 31 scene classes, each containing 60 images with 256 × 256 pixels. These images are all extracted from Google Earth.

B. Training Details

1) Experimental Parameter Settings: All experiments are implemented on Pytorch and MindSpore platforms using a GTX2080Ti GPU. Here, the SGD algorithm is selected as the optimizer. The initial learning rate of SGD is set to 0.01, and cosine decay is employed as the learning rate decay strategy. The batch size is set to 32. All models are trained for 100 iterations. The parameters pretrained on ImageNet [6] are loaded as the initial parameters of our backbone network, ResNet50 [5], then other parts of CGFNet adopt a random initialization strategy. All input images are resized to 224 × 224. The image augmentation techniques adopted are random horizontal flip and rotation.

All images of the dataset are randomly assigned to either the training set or the test set. Moreover, following previous methods [1], [9], [13], [14], [36], [37], we adopt two different settings for AID and NWPU datasets for a fair comparison. To be specific, the training-testing ratios of UCM, AID, NWPU, and OPTIMAL-31 are set to 80%–20%, 20%–80% and 50%–50%, 10%–90% and 20%–80%, and 80%–20%, respectively. We follow the corresponding reference papers to

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set the hyperparameters for comparison. The reduction ratio r in AGAB is set to 16 except 8 for the OPTIMAL-31 dataset. For FA-Loss, β in (10) is set to 3 except 4 for the AID dataset. Notably, we report the average results and standard deviations of five replicate experiments.

2) Evaluation Metrics: As one of the most widely used evaluation metrics, overall accuracy (OA) is first used to evaluate the effectiveness of our method. OA reflects the overall performance of the classification method by calculating the number of successfully classified samples divided by the total number of all test samples. Moreover, the confusion matrix (CM) is utilized to evaluate our method. In the CM, the number of correct and incorrect predictions is summarized using count values and broken down by each class. Rows denote true classes and columns denote predicted classes. CM can reveal the degree of misclassification and confusion between classes.

C. Comparison With SOTA Methods

To comprehensively evaluate the performance of the proposed CGFNet, we compare CGFNet with some SOTA approaches on four datasets (UCM, AID, NWPU, and OPTIMAL-31). In particular, the backbone networks adopted by these methods are listed in the results.

1) Results on the UC-Merced Land Use Dataset (UCM): Table I shows the OA results of different methods on UCM. It is obvious that:

1) The proposed CGFNet can achieve 99.76%, surpassing all compared methods. Particularly, the accuracy is improved by 0.54% compared to the baseline (ResNet50 [5]). This reflects the effectiveness of CGFNet for accurate aerial scene classification.
2) Some methods (e.g., SCCov [7], GBNet+global feature [8], EFPN-DSE-TDFF [38], ACR-MLFF [37], and DFAGCN [9]) obtain finer feature maps by aggregating features at different levels. Experimental results demonstrate that these methods perform better than conventional CNNs (e.g., GoogLeNet [39]). In contrast, GGAM in CGFNet shifts the model's attention to important areas of the image to enhance the feature representation. Compared with methods based on multiscale features, attention-based methods like CGFNet avoid the problems of information loss and semantic feature overlap during multiscale feature fusion.
3) The performance of ViT-based methods is comparable to CNN-based methods. Notably, the advantage of ViT-based methods is to use the self-attention mechanism to model the global context. This reflects the importance of global context to DNNs in classifying aerial scenes. It also indirectly proves the rationality of our method. That is, an adaptive pooling operation is proposed to better model the global context in aerial scenes.
4) Compared with the best methods for gathering multilevel features (ACR-MLFF [37]) and the best ViT-based method (T2T-ViT-12 [16]), the OA results are improved by 0.39% and 0.66%. Compared with EMTCAL [13], CGFNet still achieved a 0.19% improvement. The above results demonstrate the excellent performance of our proposed method on small-scale datasets.

TABLE I
EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON THE UCM DATASET. 80% OF THE DATA ARE USED FOR TRAINING

Method	Backbone	OA (%)
GoogLeNet [39]	GooLeNet	94.31±0.89
VGGNet-16 [40]	VGG	95.21±1.20
SCCov [7]	VGG	99.05±0.25
GBNet+global feature [8]	VGG	98.57±0.48
EFPN-DSE-TDFF [38]	ResNet	99.14±0.22
ACR-MLFF [37]	ResNet	99.37±0.15
DFAGCN [9]	GCN	98.48±0.42
ViT-B-16 [15]	ViT	99.28±0.23
T2T-ViT-12 [16]	ViT	99.10±0.32
PVT-V2-B0 [17]	ViT	98.86±0.38
EMTCAL [13]	ViT+ResNet	99.57±0.28
MBLANet [14]	ResNet	99.52±0.23
ResNet50+CBAM [24]	ResNet	99.28±0.24
ResNet50+EAM [41]	ResNet	98.98±0.37
Pretrained ResNet50 [5]		99.22±0.34
CGFNet (ours)		**99.76±0.23**

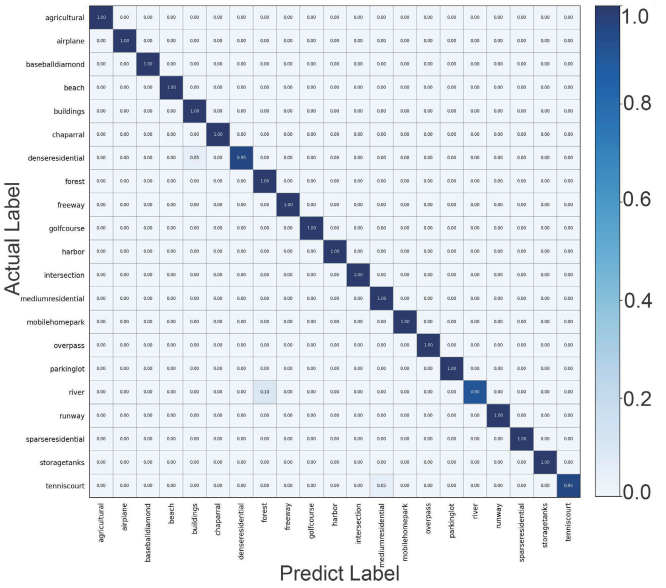


Fig. 6. CM when 80% of the data in UCM is used for training and OA equals 99.76%.

Apart from OA, Fig. 6 reports the CM of CGFNet on UCM. It can be found that most scene categories are correctly distinguished. The confusion mainly appeared in "River" and "Forest." The promising results discussed above once again corroborate that our CGFNet is effective for small datasets.

2) Results on the AID: Table II reports the OA results of CGFNet and our compared methods on AID. It is obvious that:

1) The proposed CGFNet achieves the best classification accuracy of 95.20% and 97.10% on training data amounts of 20% and 50% of AID, respectively. The OA results of CGFNet are increased by 0.73% and

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TABLE II

EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON AID DATASET. 20% AND 50% OF THE DATA ARE USED FOR TRAINING RESPECTIVELY

Method	Backbone	Traning ratio	
		20%	50%
GoogLeNet [39]	GoogLeNet	83.44±0.40	86.39±0.55
VGGNet-16 [40]	VGG	86.59±0.29	89.64±0.36
SCCov [7]	VGG	93.12±0.25	96.10±0.16
GBNet+global feature [8]	VGG	92.20±0.23	95.48±0.12
EFPN-DSE-TDFF [38]	ResNet	94.02±0.21	94.50±0.30
ACR-MLFF [37]	ResNet	92.73±0.12	95.06±0.33
DFAGCN [9]	GCN	-	94.88±0.22
ViT-B-16 [15]	ViT	93.81±0.21	96.08±0.14
T2T-ViT-12 [16]	ViT	94.39±0.22	96.29±0.24
PVT-V2-B0 [17]	ViT	93.52±0.35	96.27±0.14
EMTCAL [13]	ViT+ResNet	94.69±0.14	96.41±0.23
MBLANet [14]	ResNet	94.62±0.19	96.56±0.24
ResNet50 + CBAM [24]	ResNet	94.46±0.14	96.41±0.38
ResNet50 + EAM [41]	ResNet	94.26±0.11	97.06±0.19
Pretrained ResNet50 [5]		94.47±0.18	96.20±0.24
CGFNet (ours)		**95.20±0.14**	**97.10±0.10**

0.90%, respectively, compared with the baseline (i.e., ResNet50 [5]). The better results confirm that our method indeed guides the backbone to classify different aerial scene instances more accurately.

2) An interesting observation is that the performance of attention-based methods (e.g., DFAGCN [9] and MBLANet [14]) is significantly stronger than the backbone network of these methods. Concretely, DFAGCN [9] introduces attention mechanisms and a global cycling network during model training, which is beneficial for capturing complicated information in aerial scenes. Additionally, MBLANet embeds the newly devised module (CLAM) into ResNet50, which helps to highlight main regions on complex backgrounds and enrich feature representations. The positive experimental results confirm the vital role of attention mechanisms in enhancing the feature representation of aerial images. We develop GGAM to adaptively learn and selectively focus on important information of input images. However, we not only focus on the important information of each feature but also devise FA-Loss to optimize the feature space as a whole. Notably, our CGFNet is 2.22% higher than DFAGCN [9]. Compared with MBLANet [14], OA results are increased by 0.58% and 0.54%, respectively. Therefore, we believe that more discriminative features can be obtained through the joint action of attention mechanisms and constrained attention vector space.

Fig. 7 shows the CM for the AID dataset at 20% and 50% training ratios. CGFNet can identify most scenes, and some scenes can reach 100%, such as "Beach" and "Viaduct." However, the classification accuracy is less than 90% for some scenes (e.g., "Resort," "Center"). Further analysis found that most of them contain similar geographical features, such as "Resort" and "Park," both contain similar landscape trees.

3) Results on the NWPU RESICS-45 Dataset (NWPU): NWPU contains a great deal of scene data and is one of the most popular large-scale open datasets [4]. In the experiment, the training ratios were set to 10% and 20%, respectively. Table III presents the OA results of different methods on NWPU. It is obvious that:

1) The proposed method yields higher accuracies than other compared methods. Notably, attention-based methods (e.g., ACNet [36] and MBLANet [14]) achieve better results than traditional CNN methods. The encouraging results prove the effectiveness of using attention mechanisms to capture more important regions in this dataset.
2) For the ViT-based methods (e.g., ViT [15], EMTCAL [13], T2T-ViT-12 [16], and PVT-V2-B0 [17]), the original ViT obtained better results than classic CNNs. This shows that the global modeling ability of ViT is of great significance in NWPU. The proposed CGFNet

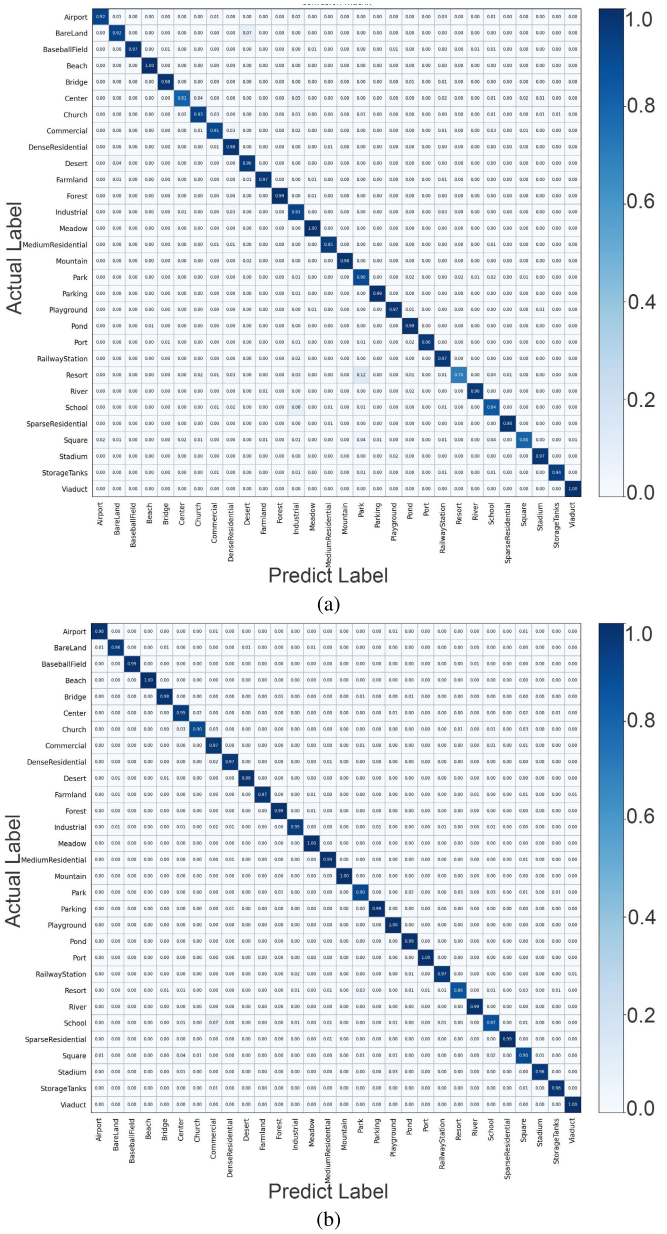


Fig. 7. CM when 20% and 50% of the data in AID are used for training and OA equals 95.23% and 97.18%, respectively. Training data accounted for (a) 20% and (b) 50%.

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TABLE III

EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON NWPU DATASET. 10% AND 20% OF THE DATA ARE USED FOR TRAINING RESPECTIVELY

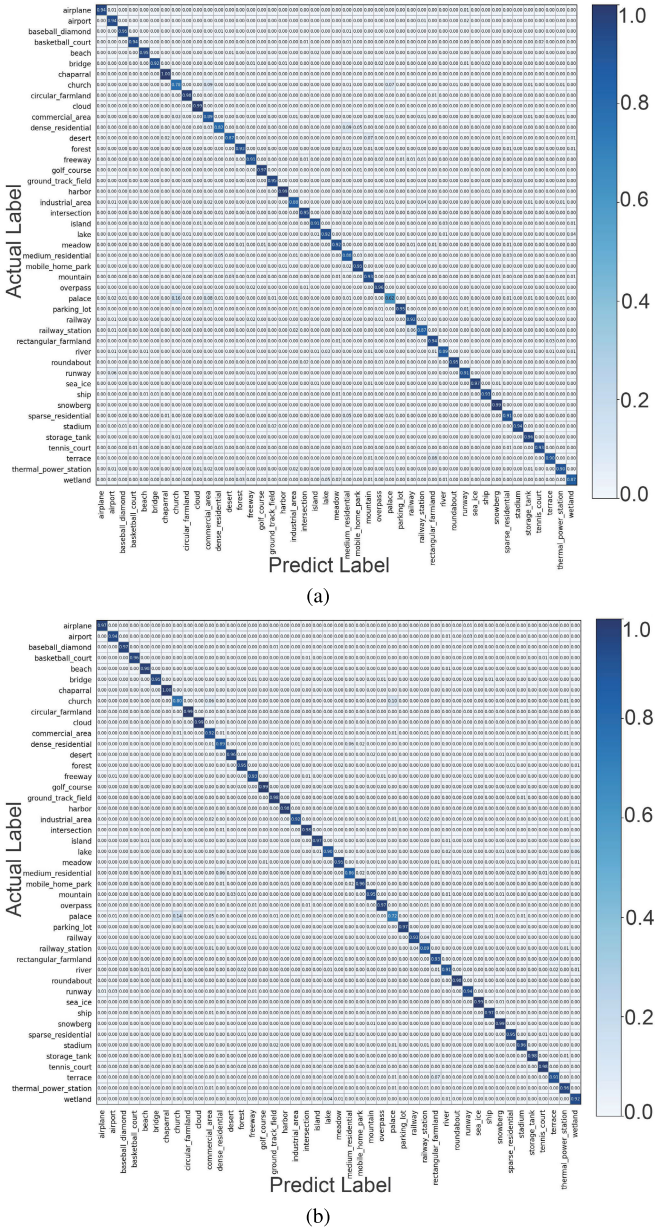
Method	Backbone	Traning ratio	
		10%	20%
GoogLeNet [39]	GoogLeNet	76.19±0.38	78.48±0.26
DFAGCN [9]	GCN	-	89.29±0.28
VGGNet-16 [40]	VGG	76.47±0.18	79.79±0.15
SCCov [7]	VGG	89.30±0.35	92.10±0.25
ACR-MLFF [37]	ResNet	90.01±0.33	92.45±0.20
ACNet [36]	VGG	91.09±0.13	92.42±0.16
ViT-B-16 [15]	ViT	90.96±0.08	93.36±0.17
T2T-ViT-12 [16]	ViT	90.62±0.18	93.19±0.10
PVT-V2-B0 [17]	ViT	89.72±0.16	92.95±0.09
EMTCAL[13]	ViT+ResNet	91.63±0.19	93.65±0.12
MBLANet [14]	ResNet	91.93±0.20	94.33±0.01
ResNet50 + CBAM [24]	ResNet	91.74±0.12	94.11±0.08
ResNet50 + EAM [41]	ResNet	91.91±0.22	94.29±0.09
Pretrained ResNet50 [5]		91.69±0.13	93.79±0.11
CGFNet (ours)		**92.24±0.22**	**94.42±0.10**

achieves better results than other ViTs, which shows its effectiveness in making up for the shortcomings of CNN in global context modeling.

3) When the training ratio is 10%, our CGFNet reaches 92.24%, 16.05% than GoogLeNet [39], 0.55% higher than ResNet50 [14], 0.31% higher than MBLANet [14], And 0.61% higher than EMTCAL [13]. When the training ratio is 20%, our CGFNet reaches 94.42%, 15.94% higher than GoogLeNet [39], 0.63% higher than ResNet50 [14], 0.09% higher than MBLANet [14], and 0.77% higher than EMTCAL [13]. These experimental results demonstrate the effectiveness of our method on large-scale datasets. It is worth noting that MBLANet also adopts the idea of using spatial attention as a supplement to channel attention. It proposes a dual attention mechanism to deal with aerial scene images with complex backgrounds. Experimental results show that this dual attention mechanism is superior to other methods in terms of OA results and stability under different training rates. In addition to the attention mechanism, we propose FA-Loss to promote the active role of label information in feature learning. In this way, label information can play a more active role in model training.

Furthermore, Fig. 8 shows the CM on the NWPU dataset. It can be concluded from Fig. 8(a) that when the training data accounts for 10%, of the 45 classes, the classification accuracy of 35 classes is greater than 90%. Similarly, from Fig. 8(b), when the training data accounts for 20%, 40 of the 45 classes have classification accuracy greater than 90%. As the training data increases, the accuracy of the model will be improved. However, the extremely similar geographic features between different classes make it harder for models to distinguish.

4) Results on the OPTIMAL-31 Dataset: The OA results of our compared methods on OPTIMAL-31 are listed in Table IV. It is obvious that:



(a)

(b)

Fig. 8. CM when 10% and 20% of the data in NWPU are used for training and OA equals 92.13% and 94.50%, respectively. Training data accounted for (a) 10% and (b) 20%.

1) The proposed CGFNet achieves 97.04% accuracy with a 1.08% improvement over our baseline (i.e., ResNet50 [5]). ResNet50 achieved the best results compared with other classic CNNs. When the attention mechanism is introduced, the attention-based CNNs (e.g., [14], [24], [41]) achieve better results than classic CNNs. These better results indicate that it is effective for CNNs to consider the attention mechanism. Despite the impressive performance achieved, the performance of CNN in aerial scene classification still needs to be further improved. In this article, we present a unified framework (i.e., CGFNet) to accurately classify aerial scenes from two aspects: focusing on the key information of input images and optimizing the feature space. Notably, the improvement obtained by

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TABLE IV
EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON OPTIMAL-31. 80% OF THE DATA ARE USED FOR TRAINING

Method	Backbone	OA (%)
GoogLeNet [39]	GoogLeNet	87.15±0.45
DenseNet-121 [43]	DenseNet	94.87±0.32
ViT-B-16 [15]	ViT	95.97±0.18
GBNet+global feature [8]	VGG	93.28±0.27
KFBNet-VGG16 [44]	VGG	95.12±0.13
KFBNet-DenseNet121 [44]	DenseNet	95.60±0.63
VSDNet-DenseNet-121 [45]	DenseNet	96.95±0.17
ConvMAE [42]	ViT	96.51
MBLANet [14]	ResNet	96.50±0.25
ResNet50 + CBAM [24]	ResNet	95.43±0.27
ResNet50 + EAM [41]	ResNet	96.45±0.28
Pretrained ResNet50 [5]		95.96±0.23
CGFNet (ours)		**97.04±0.20**

CGFNet is 1.61% (over ResNet50 + CBAM [24]), 0.59% (over ResNet50 + EAM [41]), and 0.54% (over MBLANet [14]).

2) ConvMAE [42] explored the Masked Autoencoder pretrained backbone for aerial scene classification. It combines the advantages of CNN in local inductive bias and ViT in global attention to achieve better performance. Even so, CGFNet still outperforms ConvMAE by 0.53%.

Fig. 9 shows the CM with a training ratio of 80% on the OPTIMAL-31 dataset. The classification accuracy of four out of 31 scene categories is lower than 90%. In particular, the classification accuracy of "Desert" is only 75%. Further analysis shows that confusion occurs because "Desert" and "Mountain" have the same texture.

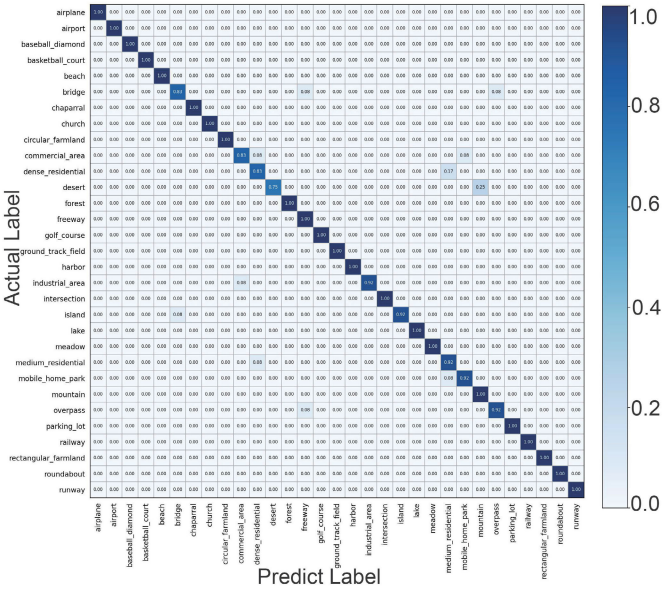


Fig. 9. CM when 80% of the data in OPTIMAL-31 is used for training and OA equals 97.04%.

D. Ablation Study

1) Design of the GGAM:

a) Adaptive global attention branch: To analyze the reduction ratio r in AGAB, we conducted ablation experiments to observe trends in model performance when the value of r changes. As shown in Fig. 10, we set r to 1, 2, 4, 8, 16, 32, and 64. Obviously, as the value of r increases, the OA results show a trend of first increasing and then decreasing. We argue that the reduction ratio r is related to the amount of information in the relationship between channels. Important information will be lost as the value of r is too large. From the experimental results, the model achieves the highest accuracy when r is set to 16, except 8 for the OPTIMAL-31 dataset.

b) Region-aware attention branch: To explore the effect of kernel size in RAAB, we employ four different sizes of kernels and compare their impact on model performance. In the experiment, AGAB is still kept because our final goal is to use both branches together. As shown in Fig. 11, we can observe that using a kernel size of 3 achieves better accuracy. In general, a larger receptive field is advantageous for determining spatially significant regions. However, due to the need to keep the original size of the feature map, an excessively large receptive field will cause the padding operation to introduce noise. At last, the kernel size of the convolutional layer is set to 3 to fuse the two attention maps in RAAB.

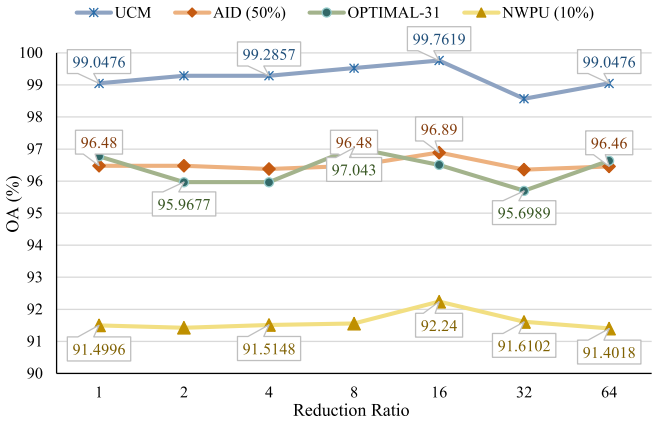


Fig. 10. Comparison of the impact of changes in reduction ratio r on different datasets.

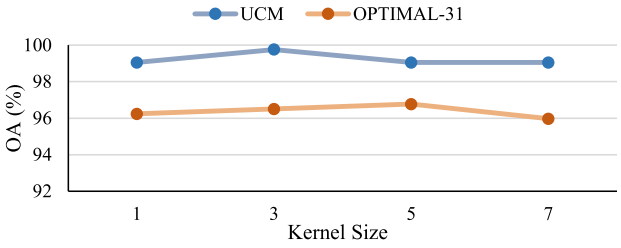


Fig. 11. Ablation study on the impact of kernel size in convolutional layers on model performance.

c) Combination of AGAB and RAAB: First, we explore two different combing strategies: sequential channel-spatial and parallel use of both attention branches. Each branch serves a different function, so the order in which they are combined may affect overall performance. From the perspective of the receptive field, AGAB introduces global contextual

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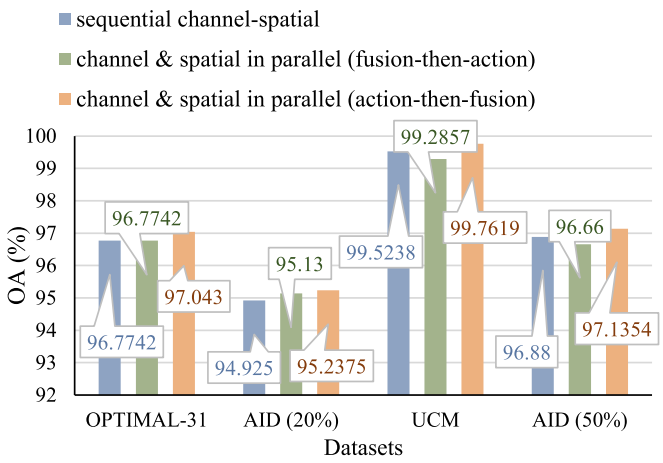


Fig. 12. Ablation study of the effect of different attention arranging strategies.

TABLE V
EVALUATION OF THE BACKBONE NETWORK OF THE MODEL

Backbones	UCM	AID (20%)	OPTIMAL-31
DenseNet201 [43]	89.76	73.92	75.54
ResNet50 [5]	99.68	95.15	97.04
MobileNetv3 [46]	98.81	92.76	94.08
ShuffleNetv2 [32]	99.28	93.69	93.82

information, while RAAB focuses on local regions by introducing grouping features into spatial attention. In the case of parallel use, we further explore the impact of the fusion order of the two branches. In particular, the fusion-then-action approach requires summing the outputs of the two branches and normalizing them using the sigmoid function. Fig. 12 plots the experimental results for three attention combination strategies. From the results, it is obvious that the attention maps inferred in parallel are more suitable for recognizing aerial scene images. Furthermore, the result of directly applying the attention map to the original feature map and then fusing them is the best of the three methods. We argue that the action-then-fusion approach enables the network to preserve information from two complementary branches without any loss of information. Note that all three different combination strategies outperform the baseline, showing that exploiting each branch is crucial, and the optimal combination strategy further improves performance.

2) Selection of Backbone: In this experiment, we explore four different CNNs as the backbone of our model: ResNet50 [5], DenseNet201 [43], MobileNetv3 [46], and ShuffleNetv2 [33]. As shown in Table V, we list the overall accuracies of different networks on three datasets. The results show that under the same hyperparameter settings, ResNet50 achieves the best results. Eventually, considering that ResNet50 has been widely used and its milestone significance, we choose ResNet50 as the backbone of our model.

3) Effectiveness of Each Branch: To validate the effectiveness of each part in CGFNet, ablation experiments are performed on the effect of each part. To be specific, the pretrained ResNet50 is employed as our backbone, and each branch is activated sequentially. As shown in Fig. 13, we can

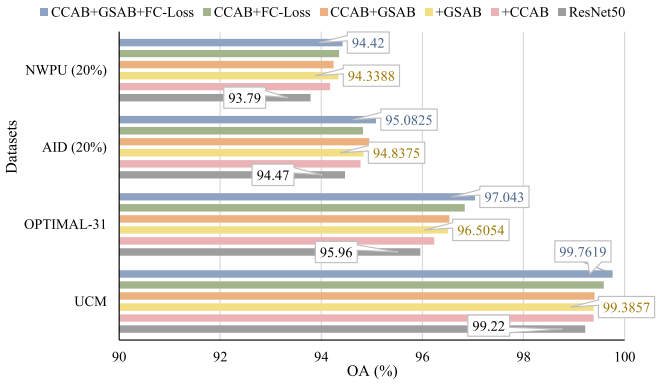


Fig. 13. Ablation study of the influence of each part in CGFNet.

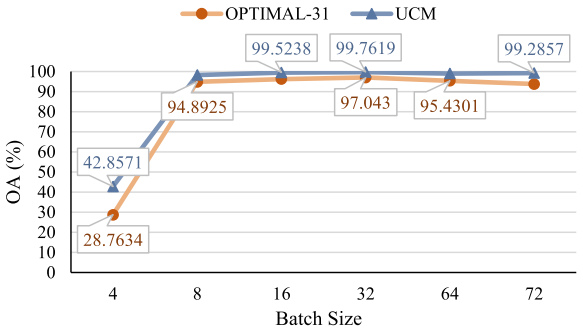


Fig. 14. Ablation study of the batch size for the FA-Loss.

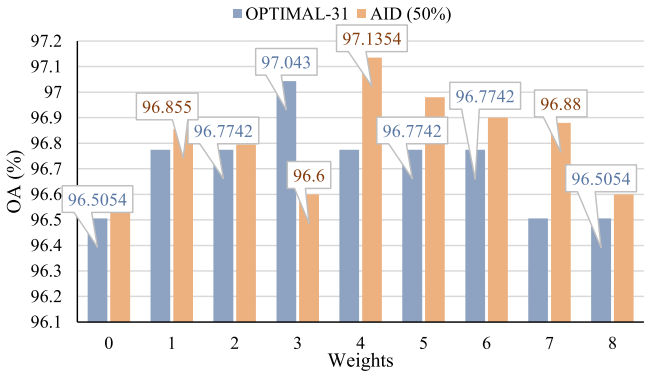


Fig. 15. Ablation study of β in (10) for the FA-Loss.

observe that all enhanced strategies outperform the original ResNet50, which shows that each branch is important and used together can further improve performance.

4) Hyperparameter Tuning on Batch Size: FA-Loss calculates the channel attention distance of each sample in a mini-batch, so the value of the batch size is of significance. We analyze the impact of batch size on model performance with ResNet50 as our backbone. As illustrated in Fig. 14, if the batch size is too small, the class relationship will disappear and the model performance will drop sharply. In this article, we empirically set the batch size to 32.

5) Effect of Weights of FA-Loss: The hyperparameter β in (10) is adopted to control the importance of FA-Loss. As shown in Fig. 15, we evaluate CGFNet with different $\beta \in \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$. The best performance is obtained

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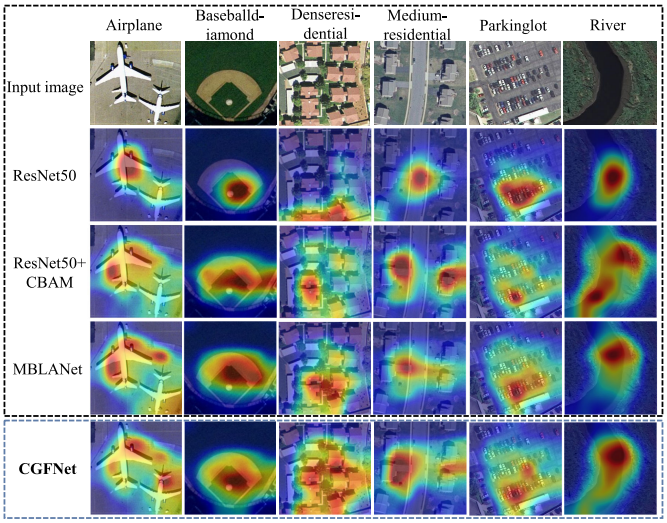


Fig. 16. Models are visualized using Grad-CAM++. The models used for comparison with CGFNet are: the baseline (ResNet50), ResNet50 + CBAM [24], and MBLANet [14]. Grad-CAM++ visualizes the features extracted by the last layer of the model.

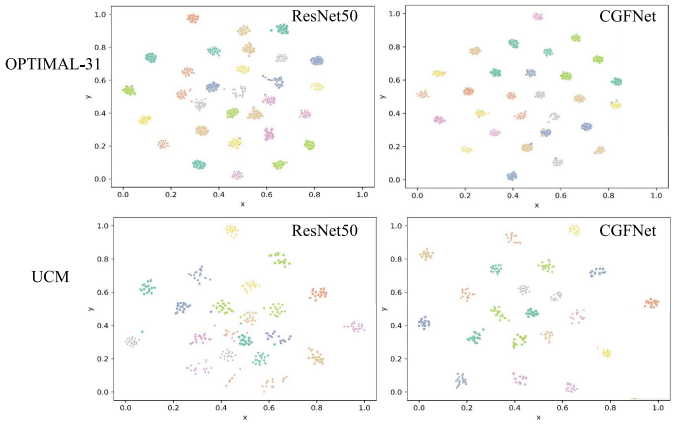


Fig. 17. Comparison of visualization using t-SNE. The results of the baseline (ResNet50) and our proposed CGFNet on two test sets are shown.

when β is set to 3 on the OPTIMAL-31 dataset and 4 on the AID dataset. It can be seen from Fig. 15 that as the value of β is gradually increased, the model accuracy shows a trend of first increasing and then decreasing. More specifically, as the value of β increases, the weight of FA-Loss also increases. FA-Loss optimizes the feature space by introducing class information, thereby improving model performance. However, blindly increasing the value of β will impair the performance since we need the correct class information provided by true labels.

E. Visualization

For qualitative analysis, Grad-CAM++ [47] is used to visualize the last layer of the model, where image samples are from the OPTIMAL-31 and UCM test sets. Grad-CAM++ is an optimized version of Grad-CAM. It can provide a visually clear explanation of deep model predictions, especially better region localization. In this experiment, we attempt to determine whether the model can learn image features well by observing the regions that the network focuses on. Fig. 16 shows the visualization results of the baseline (ResNet50), ResNet50 + CBAM [24], MBLANet [14], and CGFNet. As shown in Fig. 16, it can be found that our proposed CGFNet covers the key regions better than other methods. To put it another way, CGFNet can pay more attention to real discriminative information.

Furthermore, we employ the t-SNE algorithm [48] to visualize the model on the UCM and OPTIMAL-31 test sets. T-SNE reduces high-dimensional data to two or three dimensions to visualize high-dimensional data. It is currently the best data dimensionality reduction and visualization method. We first extract the features from the last layer of the model, and then use t-SNE for visualization to present the feature space obtained by CGFNet and ResNet50. As shown in Fig. 17, compared to the baseline, the samples in each category of our CGFNet are more clustered. This more intuitively confirms that our method can extract more discriminative features, so that the aerial scene classification task can benefit from it.

V. Conclusion

In this article, we propose to improve aerial scene classification in a unified framework (i.e., CGFNet) from two aspects: focusing on the key information of the input image and optimizing the feature space. To be specific, we propose a GGAM to adaptively learn and selectively focus on important information from input images. GGAM is inserted into each residual unit in ResNet50, which consists of two parallel branches: the AGAB and the RAAB. AGAB utilizes an adaptive pooling operation to better model the global context in aerial scenes. As a supplement to AGAB, RAAB combines grouping features with spatial attention to spatially enhance the semantic distribution of features (i.e., selectively focus on effective regions of features and ignore irrelevant semantic regions). In parallel with GGAM, a FA-Loss is exploited to introduce class information into attention vector space to indirectly optimize the feature space. Experiments on four open datasets show that our method can achieve better accuracy than current state-of-the-art (SOTA) methods.

Essentially, the performance of DNNs depends heavily on optimal hyperparameters. The hyperparameters corresponding to different datasets are usually different, which is not conducive to the expansion of the model to other fields. For further work, we intend to extend the proposed method in the following directions. First, we will combine class information with model structure more efficiently to reduce the difficulty of optimization. Second, we will automatically search for hyperparameters in FA-Loss without increasing model parameters [49], [50]. Third, we will facilitate the model to focus on important information more efficiently by more simply decoupling channel and spatial attention [51].

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